

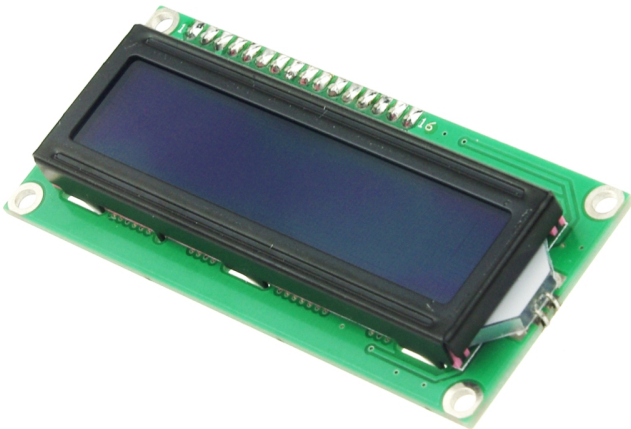
I2C TWI LCD1602 Module Gadgeteer Compatible SKU DFR0063

SKU:DFR0063 (<https://www.dfrobot.com/product-135.html>)

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Introduction

This is another great **I2C 16x2 LCD display** (<https://www.dfrobot.com/product-135.html>) compatible with Gadgeteer modules from DFRobot. With limited pin resources, your project will quickly run out of resources using normal LCDs. With this I2C interface LCD module, you only need 2 lines (I2C)to display the information.If you already have I2C devices in your project, this LCD module (<https://www.dfrobot.com/category-130.html>) actually cost no more resources at all. The adress can be set from 0x20-0x27. Fantastic for **Arduino** (<https://www.dfrobot.com/category-35.html>) or gadgeteer based projects.

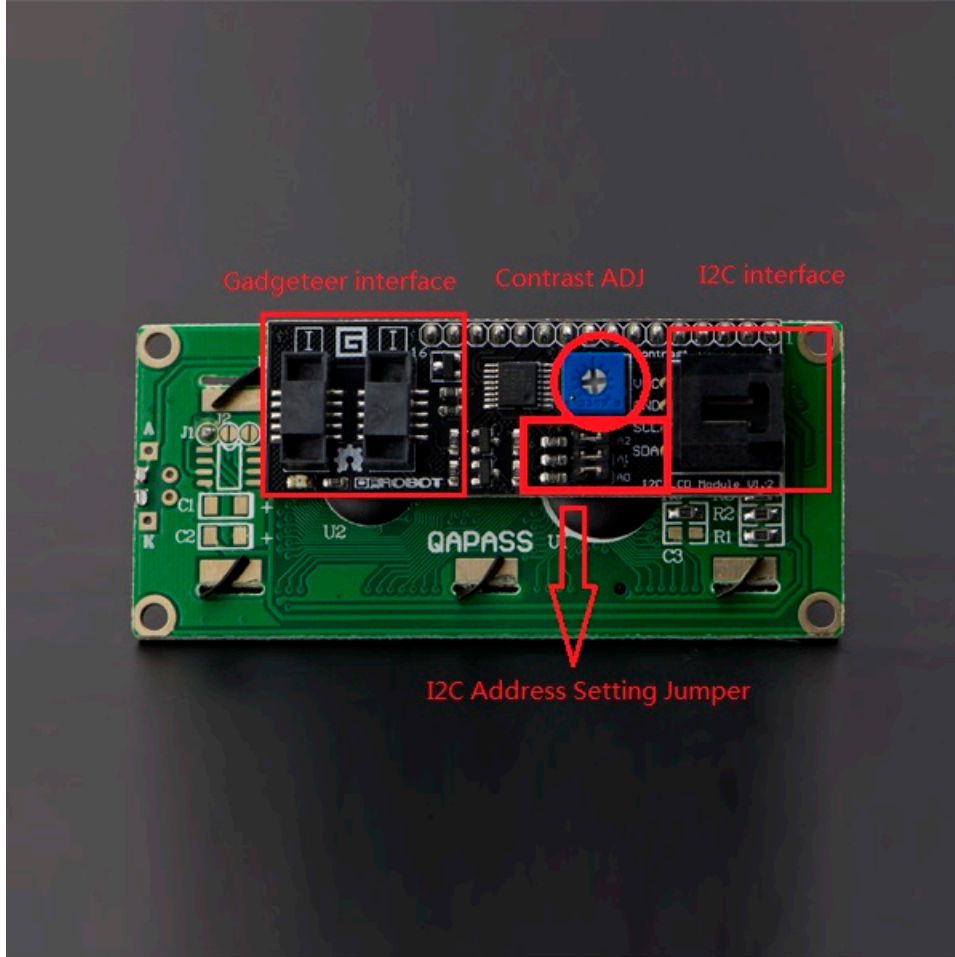


Specification

- I2C Address:0x20-0x27(0x20 default)
- Back lit (Blue with white char color)
- Supply voltage: 5V
- Interface:I2C/TWI x1,Gadgeteer interface x2
- Adjustable contrast
- Size:82x35x18 mm

Contrast Adjust

The contrast can be adjusted by the potential-meter displayed in the following picture.

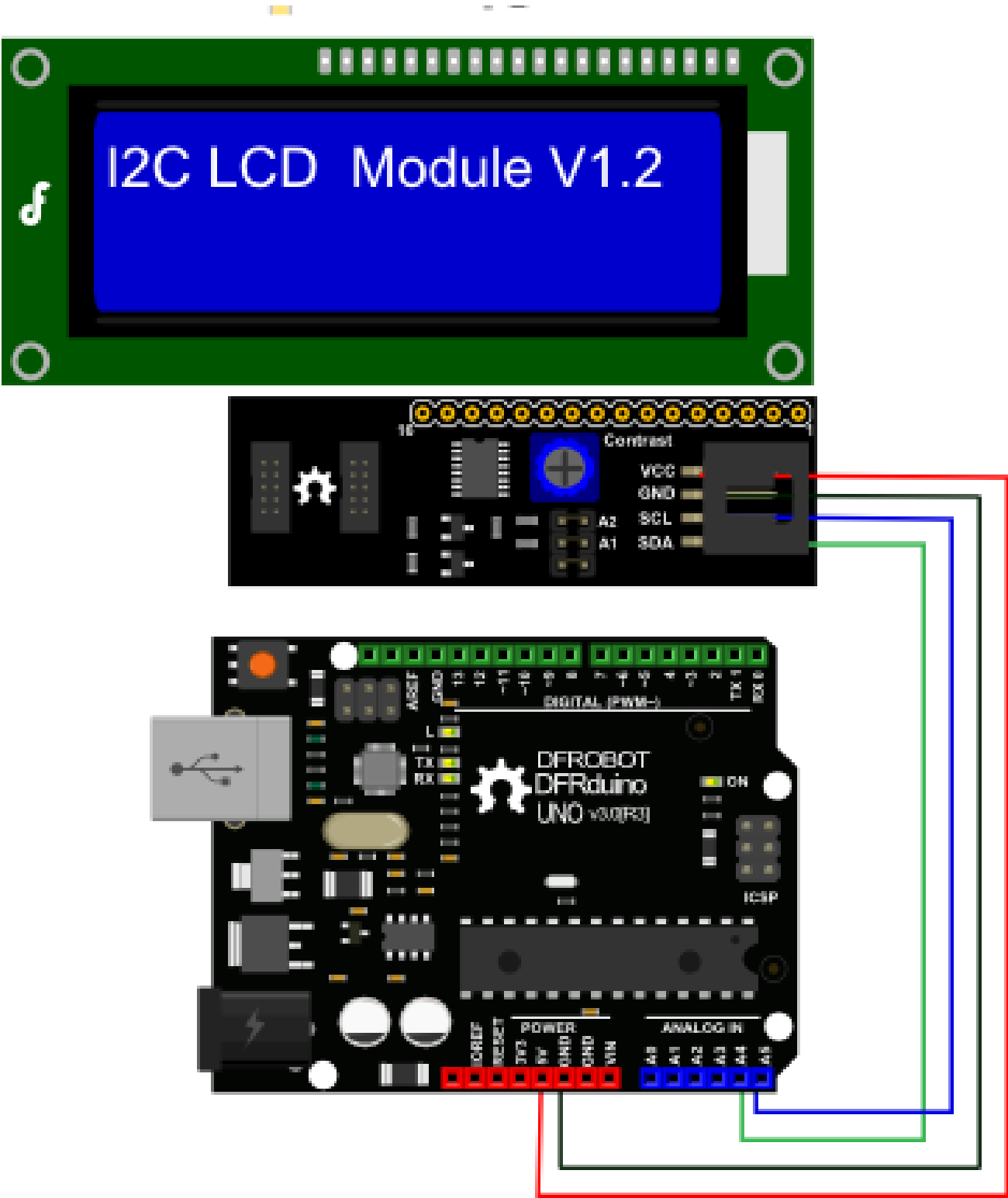


Address Setting

A2	A1	A0	IIC Address
0	0	0	0x20
0	0	1	0x21
0	1	0	0x22
0	1	1	0x23
1	0	0	0x24
1	0	1	0x25
1	1	0	0x26
1	1	1	0x27

- 0: The Jumper Cap is connected
- 1: The Jumper Cap is disconnected

NOTE: The default address is 0x20. All the jumper caps will be connected from the factory.



NOTE: V1.2 has a different power pinout from V1.1, please check the history version for the old connection diagram.

Arduino UNO (<https://www.dfrobot.com/product-610.html>): connect SDA to pin A4 and SCL to pin A5 on your Arduino.
Arduino Leonardo (<https://www.dfrobot.com/product-698.html>): connect SDA to digital pin 2 and SCL to digital pin 3 on your Arduino.

Library Support Functions

- LiquidCrystal_I2C() //set the LCD address for a 16 chars and 2 line display

- **init()** //Initialization for the LCD
- **clear()** //clear display, set cursor position to zero
- **home()** //set cursor position to zero
- **createChar()** //Fill the first 8 CGRAM locations with custom characters
- **setCursor()** //set the position of the cursor
- **cursor()** //Turns the underline cursor on
- **noCursor()** //Turns the underline cursor off
- **blink()** //Turn on the blinking cursor
- **noBlink()** //Turn off the blinking cursor
- **display()** //Turn the display on(quickly)
- **noDisplay()** //Turn the display Off(quickly)
- **backlight()** //Turn the backlight on
- **noBacklight()** //Turn the backlight off
- **scrollDisplayLeft()** //Make the display scroll left without changing the RAM
- **scrollDisplayRight()** //Make the display scroll right without changing the RAM
- **autoscroll()** //This will 'right justify' text from the cursor
- **noAutoscroll()** //This will 'left justify' text from the cursor
- **leftToRight()** //This is for text that flows Left to Right
- **rightToLeft()** //This is for text that flows Right to Left

Sample Code

Download Sample code and library

(https://raw.githubusercontent.com/DFRobot/WikiResource/master/DFR0063/LiquidCrystal_I2C.zip)

```
>
//DFRobot.com
//Compatible with the Arduino IDE 1.0
//Library version:1.1
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x20,16,2); // set the LCD address to 0x20 for a 16 chars and 2 line display

void setup()
{
    lcd.init();                      // initialize the lcd

    // Print a message to the LCD.
    lcd.backlight();
    lcd.print("Hello, world!");
}

void loop()
{
}
```

Sample sketch: Control the back light of the I2C LCD1602 Module

```

#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#if defined(ARDUINO) && ARDUINO >= 100
#define printByte(args)  write(args);
#else
#define printByte(args)  print(args,BYTE);
#endif

LiquidCrystal_I2C lcd(0x20,16,2);  // set the LCD address to 0x20 for a 16 chars and 2 line display

void setup(){

    lcd.init();                      // initialize the lcd
    lcd.backlight();

    lcd.home();

    lcd.print("Hello world...");
    lcd.setCursor(0, 1);
    lcd.print("dfrobot.com");

}

int backlightState = LOW;
long previousMillis = 0;
long interval = 1000;

void loop(){

    unsigned long currentMillis = millis();

    if(currentMillis - previousMillis > interval) {
        previousMillis = currentMillis;

        if (backlightState == LOW)
            backlightState = HIGH;
        else
            backlightState = LOW;

        if(backlightState == HIGH)  lcd.backlight();
        else lcd.noBacklight();
    }
}

```

If you want to use the library's own sample code, pay attention to modify the initialization statement, need to change:

```
LiquidCrystal_I2C lcd(0x27,16,2); // set the LCD address to 0x27 for a 16 chars and 2 line display
```

to:

```
LiquidCrystal_I2C lcd(0x20,16,2); // set the LCD address to 0x20 for a 16 chars and 2 line display(All jumpers should be conn
```

Because the default initialization statement is for LCD1602!

Old version wiki

- I2C/TWI LCD1602 Module (https://www.dfrobot.com/wiki/index.php/I2C/TWI_LCD1602_Module__SKU:_DFR0063_)

Projects

- DYP-ME007 Ultrasound range finder - display distance on a I2C 2x16 LCD (<https://forum.arduino.cc/t/dyp-me007-ultrasound-range-finder-display-distance-on-a-i2c-2x16-lcd/63060>)

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